

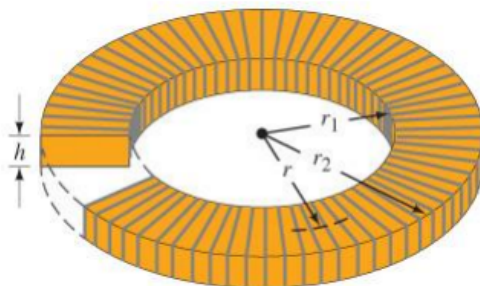
Week 13 Discussion Problems

1. Consider a long solenoid with turn density n , radius R , length ℓ , and current I .



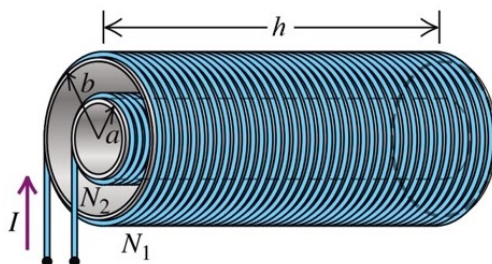
- Use Ampère's law to calculate the magnetic field B inside the solenoid.
- Use the result for a single turn, $B = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$, to calculate B on the symmetry axis. You might find the following integral useful: $\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{3/2}} = 2$.
- Determine the self-inductance of the solenoid. Hint: $L = \frac{\phi_B}{I}$ for each turn.

2. A toroidal coil of inner radius r_1 , outer radius r_2 , and height h contains N turns and carries a current I . Assume that the current passes from left to right at the top of the cross-section shown in Fig. 6.

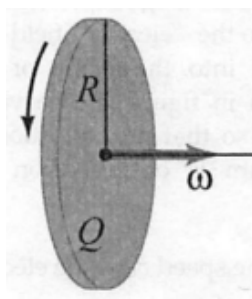


- Determine the direction and magnitude of the magnetic field created inside this toroidal coil.
- Calculate the self-inductance L using the definition $\frac{\phi_B}{I}$.
- Calculate the total energy stored in the coil. Hint: $dU = \frac{B^2}{2\mu_0}$.
- Recalculate the self-inductance using $U = \frac{1}{2}LI^2$.

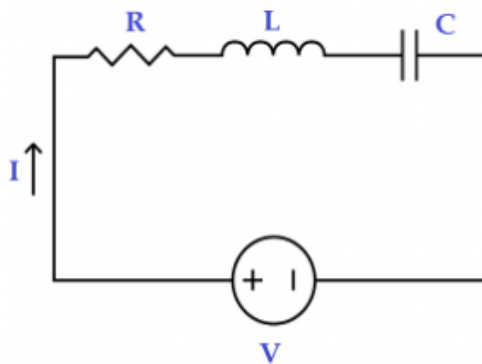
3. Consider two long coaxial solenoids with different cross-sectional areas and different turn densities. Let solenoid 1 have turn density n_1 and area A_1 , and solenoid 2 have turn density n_2 and area A_2 . Take $A_1 < A_2$.



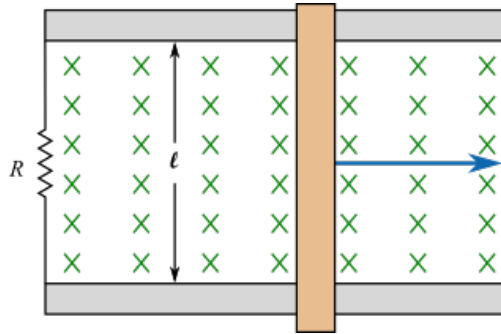
- (a) Find the magnetic field produced by solenoid 1.
 (b) Determine the magnetic flux through one turn of solenoid 2 due to the field of solenoid 1.
 (c) Compute the mutual inductance M .
 (d) Does M depend on the role of each solenoid in the calculation?
4. A non-conducting circular disk of negligible thickness and radius R carries a uniformly distributed electric charge Q . The disk spins about its symmetry axis with angular speed ω .



- (a) Determine the infinitesimal magnetic dipole moment of a thin circular ring of inner radius r and width dr . Hint: $d\vec{\mu} = I d\vec{A}$.
 (b) Integrate to obtain the total magnetic dipole moment of the spinning disk.
 (c) What is the torque exerted by an external magnetic field B at an angle θ relative to the magnetic dipole moment vector? Hint: $\vec{\tau} = \vec{\mu} \times \vec{B}$.
5. Consider a resistor R , capacitor C , inductor L , and battery with emf $\mathcal{E}$ all in series.



- (a) Use Kirchoff's loop law to find a differential equation in terms of the charge $Q(t)$ on the capacitor as a function of time.
 (b) Given $A \frac{d^2 f}{dt^2} + B \frac{df}{dt} + f = C$ has solution $f(t) = c_1 \exp\left(\frac{-B + \sqrt{B^2 - 4A}}{2A} t\right) + c_2 \exp\left(\frac{-B - \sqrt{B^2 - 4A}}{2A} t\right) + C$ for $B^2 \neq 4A$, write down an expression for $Q(t)$ satisfying $Q(t = 0) = 0$ and $I(t = 0) = 0$. What happens when (1) $RC > 4\frac{L}{R}$ and (2) $RC < 4\frac{L}{R}$?
6. A conducting bar of length ℓ slides with initial speed v_0 on frictionless conducting tracks connected by a wire of resistance R , allowing current to flow in a square loop. A uniform magnetic field B points perpendicular to the plane of the loop.



- (a) Use Faraday's law to determine the induced emf, the current in the loop, and the electrical power dissipated in the circuit.
- (b) Compute the magnetic force on the moving bar and use it to determine the mechanical power delivered against the magnetic force.
- (c) What is the total energy dissipated in the circuit by the time the bar comes to rest?